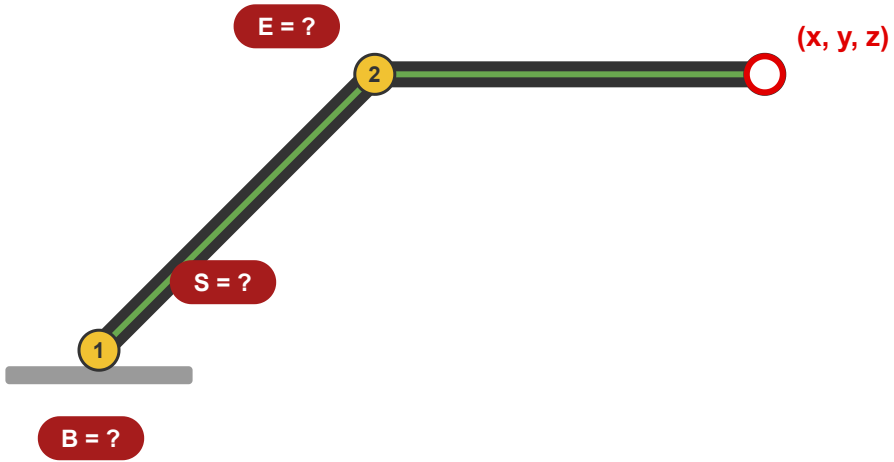
ROBOTIC ARM COURSE · LESSON 6 OF 12

The Trigonometric Approach to Inverse Kinematics

Deriving IK with the law of cosines and trigonometry

Day 1 · Solving the flat arm with triangles

Day 2 · Extending to 3D



RECAP · LESSONS 4 AND 5

Where Lesson 5 Left Off

GIVEN · THE JOINT ANGLES

B = 45°, S = 45°, E = 45°

→

FORWARD KINEMATICS GAVE

(x, y, z) = (301.78, 301.78, 176.78)

Lesson 4's matrices (and Lesson 5's code) answer one question: **"If the joints turn like this, where does the tip end up?"**

Today we ask the opposite question.

Notation reminder. z points up. B turns the base, S is the shoulder angle measured clockwise from vertical, and E is how far the elbow bends (E = 0° means a straight arm). $L_1 = L_2 = 250$ mm.

WHY IK

Forward vs. Inverse Kinematics

FORWARD KINEMATICS (FK)

angles → tip position

Given B, S, E, find (x, y, z).
Always exactly **one** answer.

INVERSE KINEMATICS (IK)

tip position → angles

Given (x, y, z), find B, S, E.
Could be **none, one, or two** answers.

A robot is given a **target**, like "pick up the cup at (x, y, z)". The motors need **angles**. IK is the step that turns one into the other.

Lesson 3 used triangles to go forward. Today we use the **same triangles backwards**.

DAY 1 · THE PROBLEM

The IK Problem (Flat Arm)

First the flat arm of Lesson 4, with the target at **(X, Y)**.

GIVEN

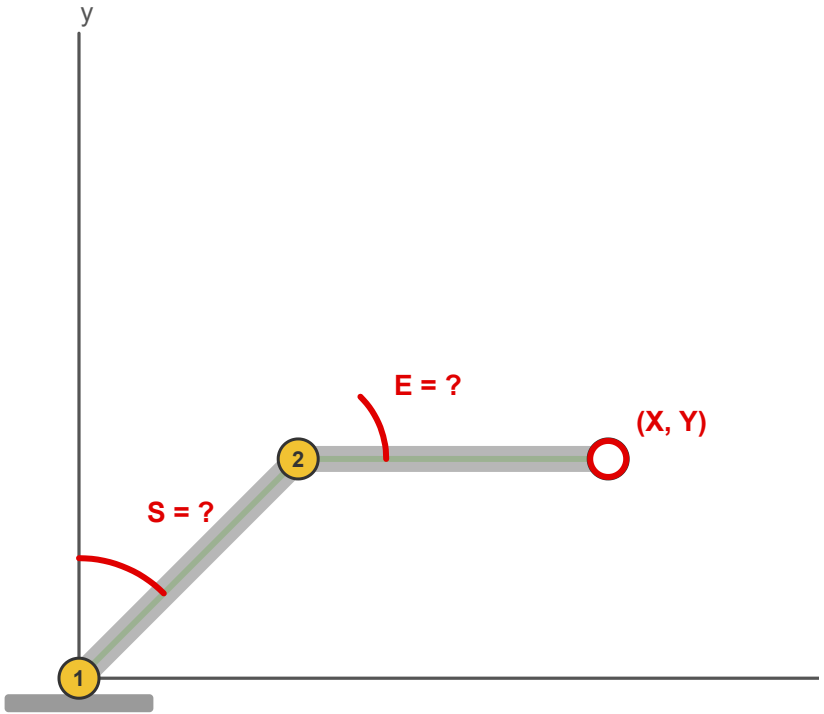
$L_1 = L_2 = 250$ mm

Target (X, Y)

FIND

S and E

Solve the flat arm first. In Day 2 the base angle B turns it into the full 3D arm.



DAY 1 · LAW OF COSINES

The Law of Cosines

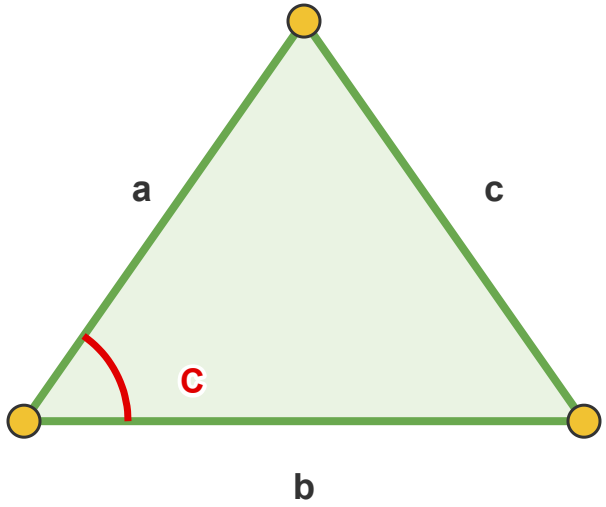
For **any** triangle with sides a, b, c, where C is the angle opposite side c:

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Solve it for the angle:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Use it when you know all three sides and want an angle. If C = 90°, cos C = 0 and it becomes Pythagoras.



DAY 1 · LAW OF COSINES

Worked Example: Law of Cosines

A triangle has sides a = 250, b = 250 and c = 300. Find angle C.

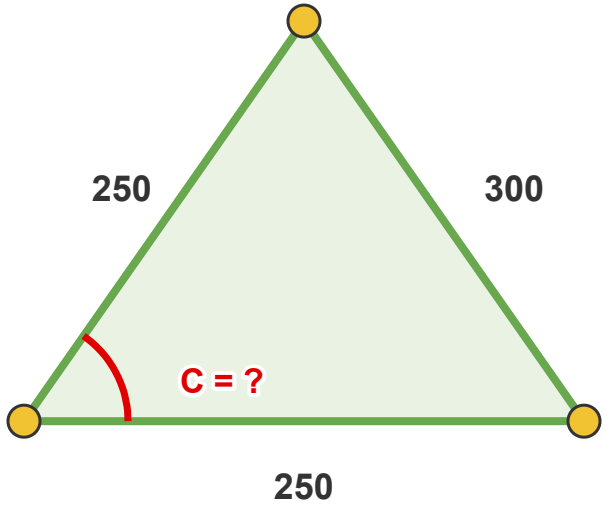
$$\cos C = \frac{250^2 + 250^2 - 300^2}{2 \times 250 \times 250}$$

$$= \frac{125{,}000 - 90{,}000}{125{,}000} = 0.28$$

UNDO THE COSINE

$C = \cos^{-1}(0.28) = \mathbf{73.74°}$

$\cos^{-1}$ is the “inverse cosine” key (arccos) on a calculator. Set it to degrees.



DAY 1 · TRY IT YOURSELF

Try It Yourself #1

Find angle C opposite side c. Both a and b are 250.

(a) c = 250 **(b) c = 400**

SOLUTION

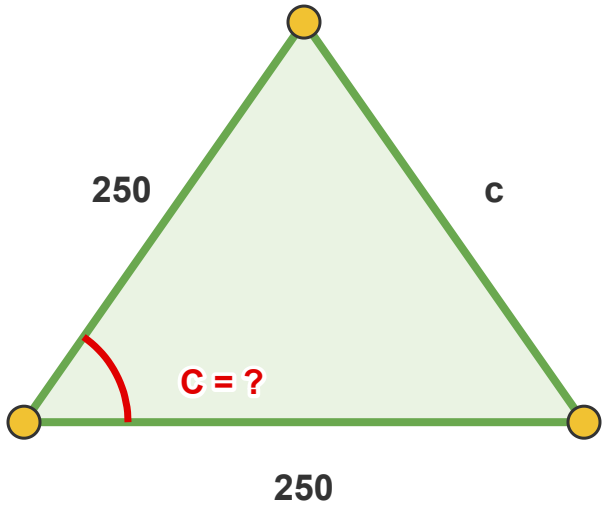
(a) cos C = (125,000 − 62,500) ÷ 125,000 = 0.5

C = 60°

(b) cos C = (125,000 − 160,000) ÷ 125,000 = −0.28

C = 106.26°

A negative cosine means an obtuse angle (more than 90°).



DAY 1 · THE ARM IS A TRIANGLE

The Arm Makes a Triangle

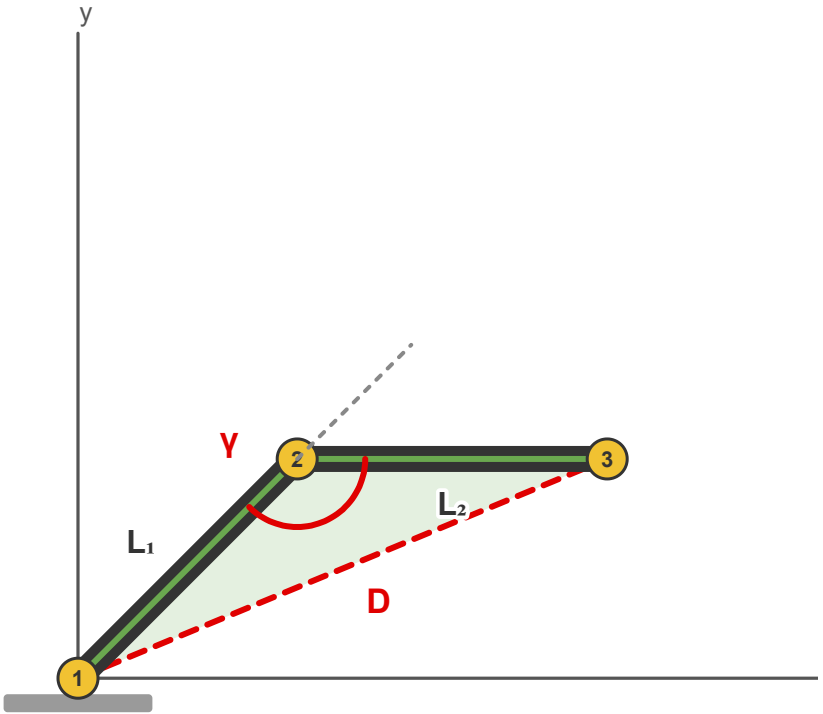
Draw a straight line D from the shoulder to the target. The two links and D form a triangle.

WE KNOW ALL THREE SIDES

$L_1 = 250 \cdot L_2 = 250$

$D = \sqrt{(X^2 + Y^2)}$ (Pythagoras)

All three sides known means the law of cosines can find the **angles**. Those angles are the joint angles.



DAY 1 · STEP 1 · THE ELBOW

Find the Elbow Angle E

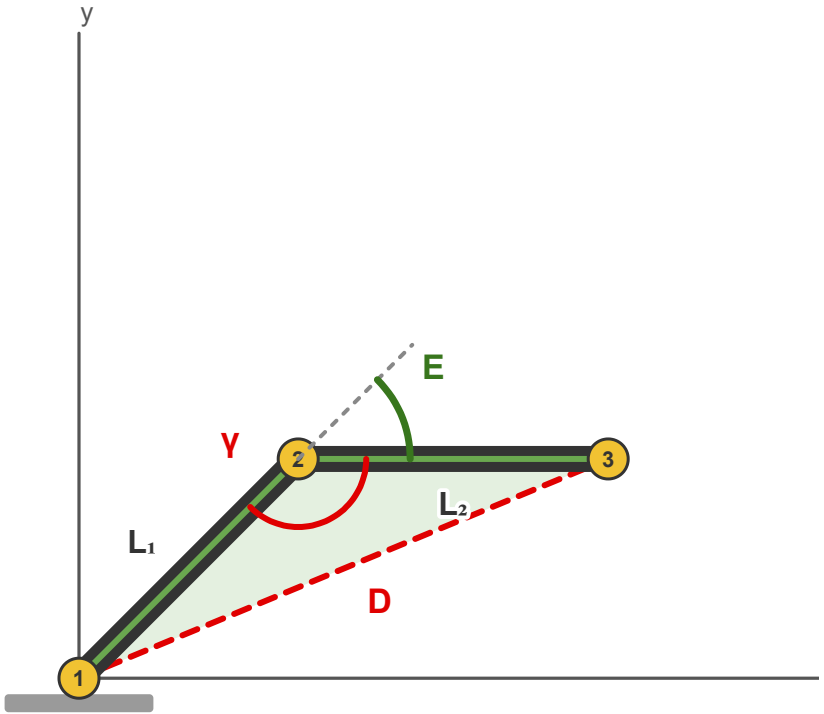
The law of cosines gives the angle **γ** *inside* the triangle (opposite D):

$$\cos\gamma = \frac{L_1^2 + L_2^2 - D^2}{2\,L_1\,L_2}$$

But E is the bend *outside* the triangle, so **E = 180° − γ**. Since cos(180° − γ) = −cos γ:

$$\cos E = \frac{D^2 - L_1^2 - L_2^2}{2\,L_1\,L_2}$$

Check: a straight arm has D = 500, so cos E = 1 and E = 0°. ✓



DAY 1 · WORKED EXAMPLE · STEP 1

Worked Example: The Elbow

Target (X, Y) = **(426.78, 176.78)**. Find E.

THE LINE TO THE TARGET

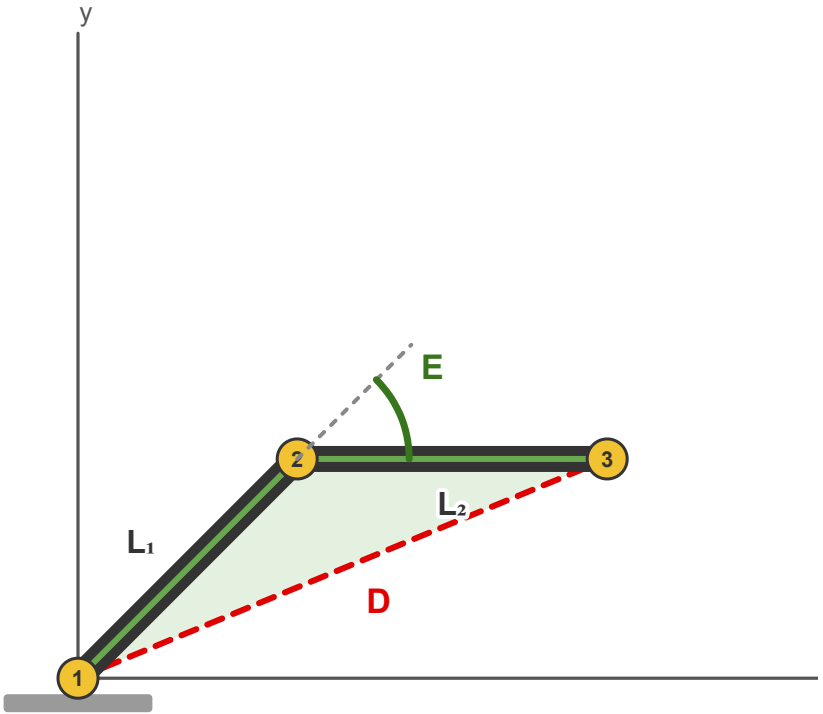
$D^2 = X^2 + Y^2 = 426.78^2 + 176.78^2 = 213{,}389$

D = 461.94 mm

THE ELBOW

cos E = (213,389 − 62,500 − 62,500) ÷ 125,000 = 0.7071

$E = \cos^{-1}(0.7071) = 45°$



DAY 1 · STEP 2 · THE SHOULDER

Find the Shoulder Angle S

The upper arm sits **two angles** above the horizontal:

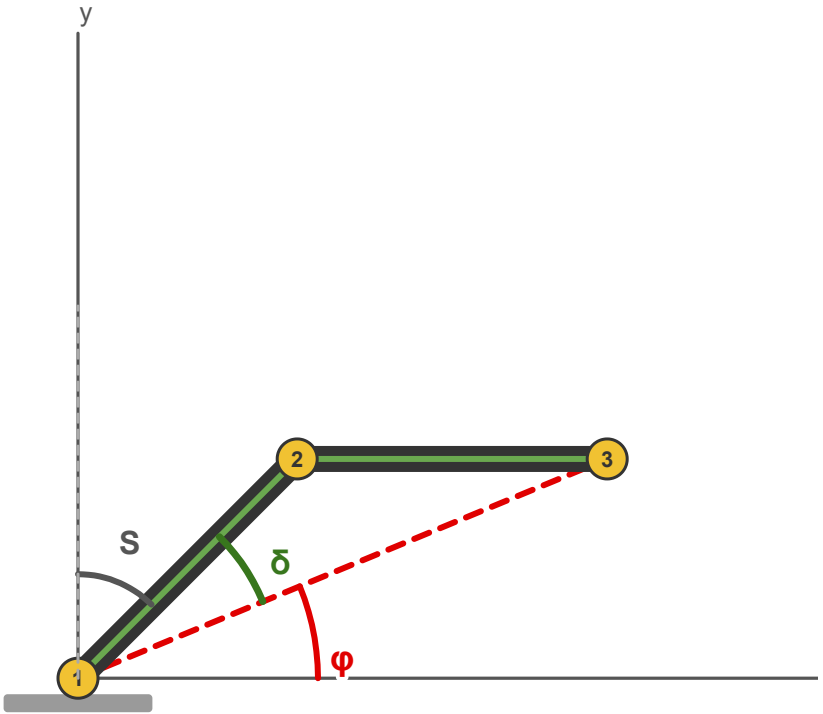
φ the tilt of the line D:
$\varphi = \tan^{-1}(Y \div X)$

δ the angle between D and the upper arm, from the law of cosines:

$$\cos\delta = \frac{L_1^2 + D^2 - L_2^2}{2\,L_1\,D}$$

Add them: **a = φ + δ**. Lesson 4 measured S from vertical, so

$$S = 90° - a = 90° - (\varphi + \delta)$$



DAY 1 · WORKED EXAMPLE · STEP 2

Worked Example: The Shoulder

Same target (426.78, 176.78), with D = 461.94 and D^2 = 213,389. Note 230,970 = 2 × 250 × 461.94.

TILT OF THE LINE

$\varphi = \tan^{-1}(176.78 \div 426.78) =$ **22.5°**

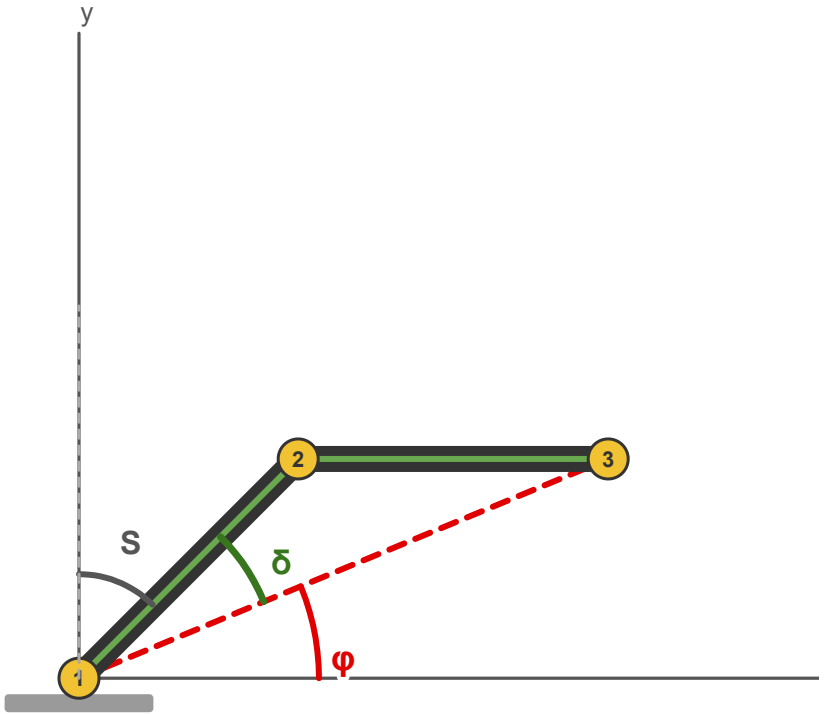
ANGLE TO THE UPPER ARM

$\cos\delta = (62{,}500 + 213{,}389 - 62{,}500) \div 230{,}970$

$= 213{,}389 \div 230{,}970 = 0.9239$

$\delta = \cos^{-1}(0.9239) =$ **22.5°**

a = 22.5° + 22.5° = 45°, so **S = 90° − 45° = 45°**.



DAY 1 · WORKED EXAMPLE

Completed Variables

$L_1 = L_2 = 250$ mm

(X, Y) = (426.78, 176.78)

D = 461.94 mm

φ = 22.5°

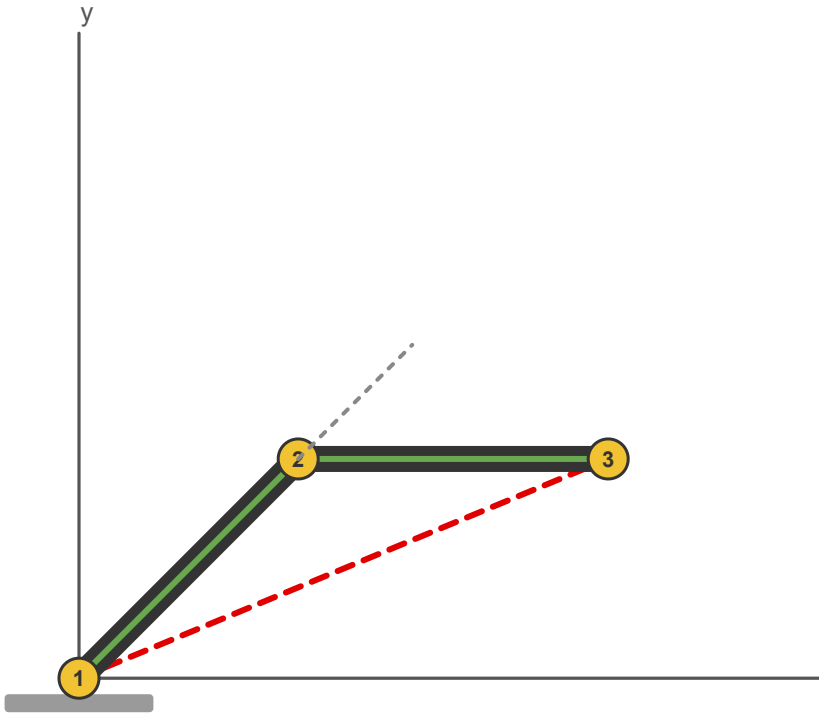
δ = 22.5°

a = 45°

E = 45°

S = 45°

IK undid FK. These are the angles Lesson 4 started with. Put S = 45° and E = 45° back into the chain from Lesson 4 and you get (426.78, 176.78) again.



DAY 1 · TRY IT YOURSELF

Try It Yourself #2

Find S and E for the target (X, Y) = **(341.51, 341.51)**.

$L_1 = L_2 = 250$ mm

SOLUTION

$D^2 = 233{,}255$ ($D = 482.96$)

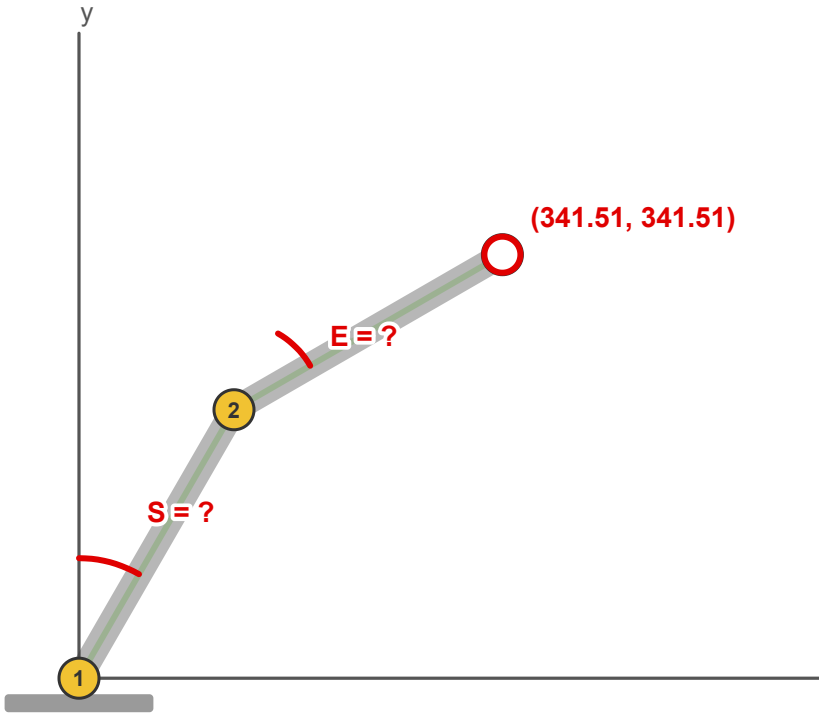
$\cos E = (233{,}255 - 125{,}000) \div 125{,}000 = 0.866$

$E = 30°$

$\varphi = \tan^{-1}(1) = 45°$, $\cos \delta = 233{,}255 \div 241{,}480 = 0.9659$, $\delta = 15°$

$S = 90° - (45° + 15°) = 30°$

The answer to Lesson 4's Try It Yourself #2, run backwards.



DAY 1 · TRY IT YOURSELF

Try It Yourself #3

Find S and E for the target (X, Y) = **(250, 250)**.

$L_1 = L_2 = 250$ mm

SOLUTION

$D^2 = 125{,}000$ ($D = 353.55$)

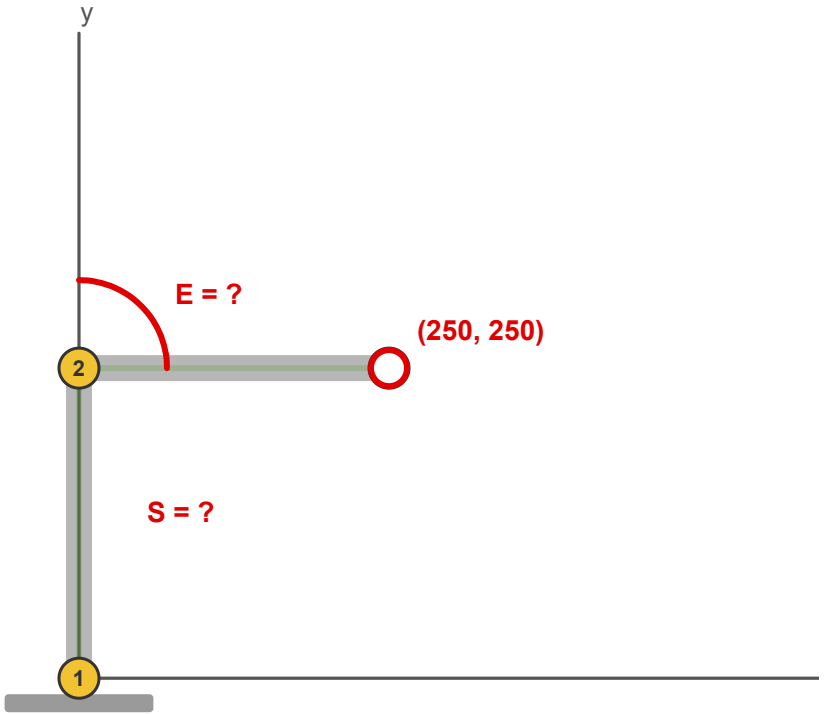
$\cos E = (125{,}000 - 125{,}000) \div 125{,}000 = 0$

$E = 90°$

$\varphi = 45°$, $\cos \delta = (62{,}500 + 125{,}000 - 62{,}500) \div (2 \times 250 \times 353.55) = 0.7071$, $\delta = 45°$

$S = 90° - (45° + 45°) = 0°$

The upper arm stands straight up and the forearm lies flat.



DAY 1 · TWO SOLUTIONS

Elbow Up or Elbow Down

One target can often be reached **two ways**: the elbow can point up or down. Both put the tip at (250, 250).

ELBOW UP · OUR DEFAULT

$E = +\cos^{-1}(\ldots)$ and **$a = \varphi + \delta$**

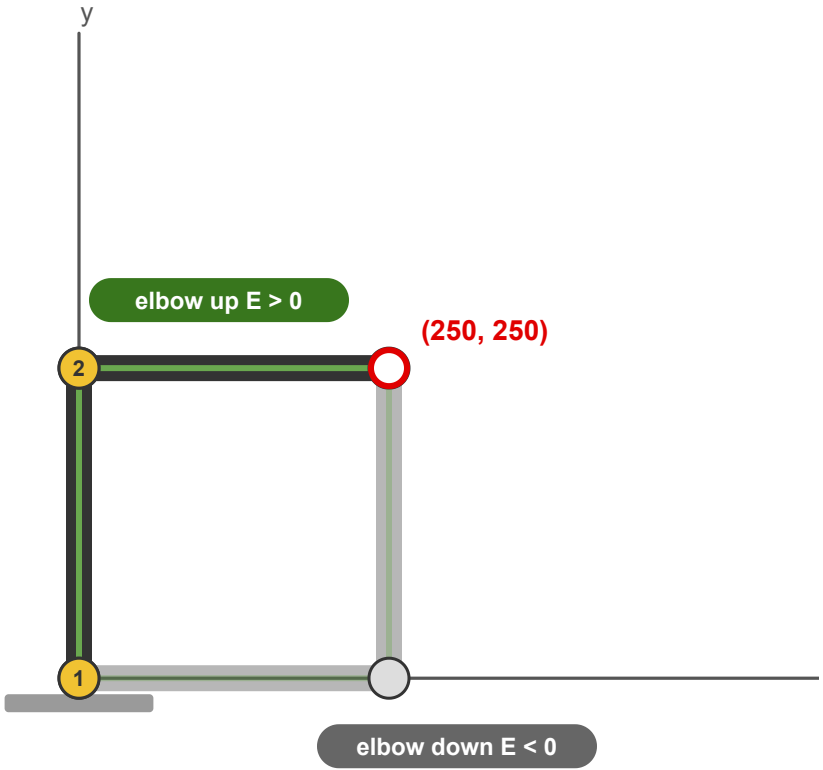
(250, 250): E = 90°, S = 0°

ELBOW DOWN

$E = -\cos^{-1}(\ldots)$ and **$a = \varphi - \delta$**

(250, 250): E = −90°, S = 90°

The arccos always gives a positive answer, so we choose the sign. Check your arm's joint limits before using elbow down.



DAY 1 · REACHABILITY

Can the Arm Reach It?

A triangle only exists if its sides fit together:

$$D \le L_1 + L_2 = 500 \text{ mm}$$

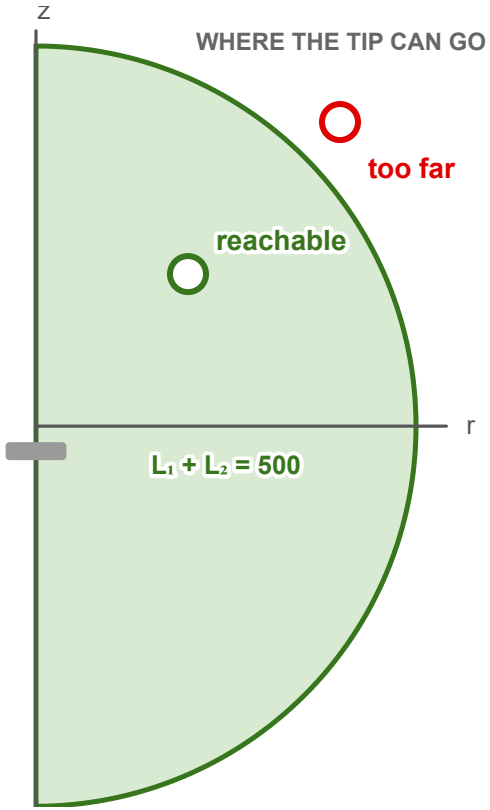
If D is bigger, the links cannot stretch far enough. The formula warns you:

THE SIGN OF TROUBLE

cos E > 1 (or < −1) is impossible, so $\cos^{-1}$ fails.

→ The target is **out of reach**.

Always check cos E first. A robot must never be told to go somewhere it cannot.



DAY 1 · RECAP

Day 1 Recap

TRIANGLE

L_1, L_2 and D

The links and the line to the target make a triangle. $D = \sqrt{(X^2 + Y^2)}$.

ELBOW

$\cos E = (D^2 - L_1^2 - L_2^2) \div 2L_1L_2$

Law of cosines. E = 0° when the arm is straight.

SHOULDER

$S = 90° - (\varphi + \delta)$

$\varphi = \tan^{-1}(Y \div X)$. δ comes from the law of cosines again.

CHECK

$D \leq 500$, and two solutions

Out of reach if $|\cos E| > 1$. Elbow up or elbow down.

Next time: the real arm's base turns too, so we need 3D.

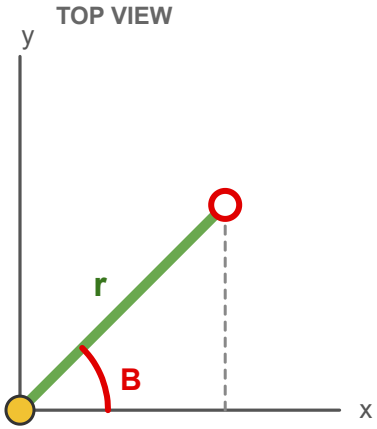
DAY 2 · EXTENDING TO 3D

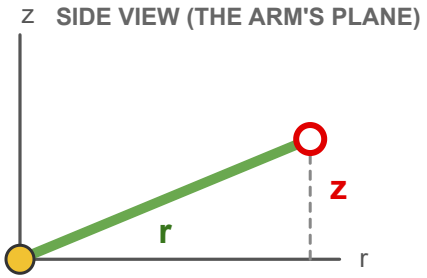
Extending to 3D

Day 1 solved a flat arm in one vertical plane. Now the target is a point **(x, y, z)** and the base turns that plane.

> **Day 1 in one line:** target (X, Y) → E, then S

The plan: use the base to point the arm's plane at the target. Then **flatten** the problem and reuse Day 1.





DAY 2 · STEP 1 · THE BASE

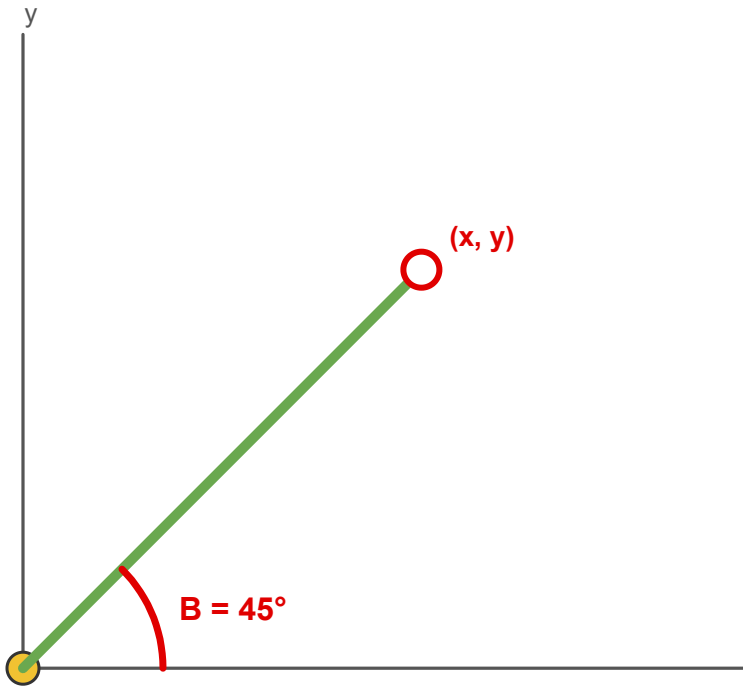
The Base Aims the Arm

Look from above. The base must turn until the arm points at the target's (x, y) spot.

$$B = \tan^{-1}(y \div x)$$

Example: the target **(301.78, 301.78, 176.78)** has x = y, so tan B = 1 and **B = 45°**.

Careful: the plain $\tan^{-1}$ can point the wrong way. The next slide shows when.



DAY 2 · STEP 2 · FLATTEN

Flatten 3D Into 2D

The arm lives in one vertical plane. Measure the target inside that plane:

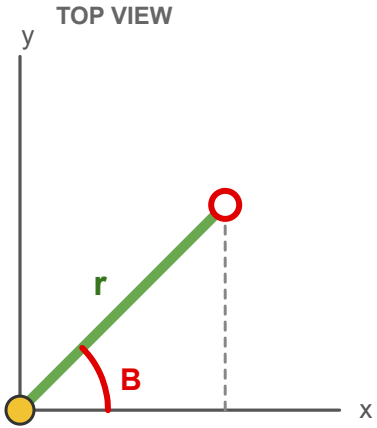
HORIZONTAL REACH

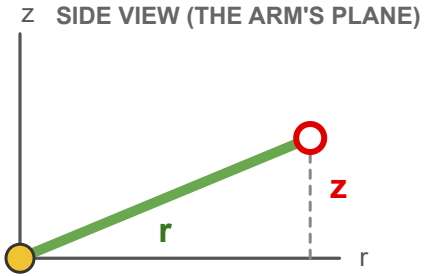
$$r = \sqrt{(x^2 + y^2)}$$

HEIGHT

$$h = z$$

Now (r, h) plays the role of **(X, Y)** from Day 1. The elbow and shoulder steps need no changes.





DAY 2 · STEP 1 · THE BASE

Why atan2?

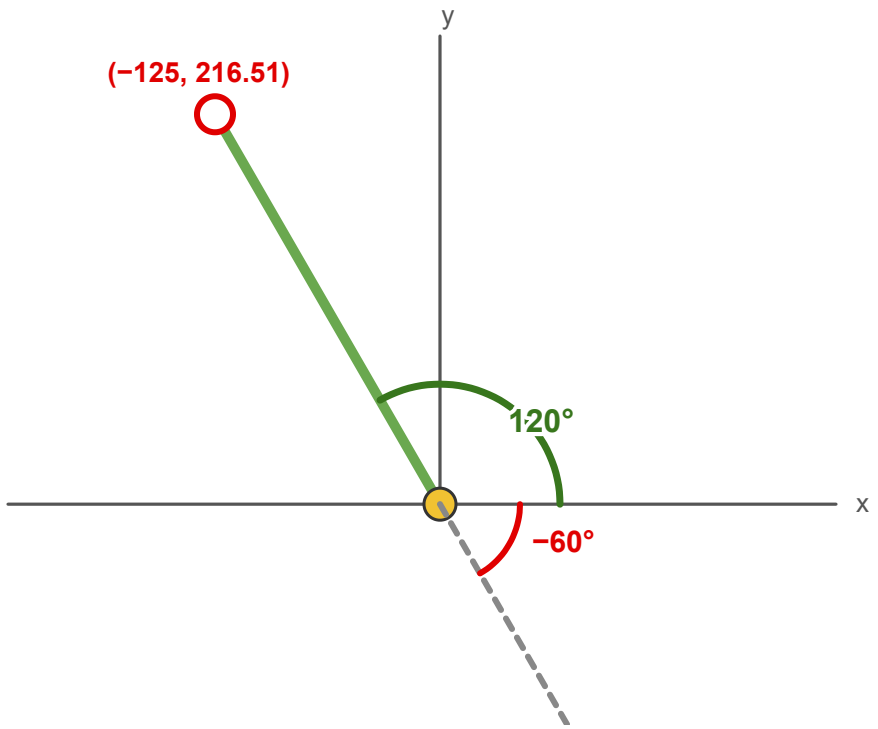
Try the target **(−125, 216.51)**:

$\tan^{-1}(216.51 \div -125) = \mathbf{-60°}$. That points the wrong way, at (125, −216.51).

ATAN2(Y, X)

Looks at the signs of **both** y and x, so it picks the right quadrant: **B = 120°**.

Python: `np.degrees(np.arctan2(y, x))`. If x = y = 0, the tip is on the base's axis and every B works (see the Edge Cases slide).



DAY 2 · THE RECIPE

The Full Recipe

1. **Aim the base**: $B = \operatorname{atan2}(y, x)$

2. **Flatten**: $r = \sqrt{(x^2 + y^2)}$, $h = z$, $D^2 = r^2 + h^2$

3. **Elbow**: $\cos E = \dfrac{D^2 - L_1^2 - L_2^2}{2\,L_1\,L_2}$ (check $|\cos E| \le 1$ first)

4. **Shoulder**: $S = 90° - (\varphi + \delta)$, $\varphi = \tan^{-1}(h \div r)$, $\cos \delta = \dfrac{L_1^2 + D^2 - L_2^2}{2\,L_1\,D}$

Four steps, a few formulas, and an exact answer. No guessing and no repeating.

DAY 2 · WORKED EXAMPLE

Worked Example: Setup

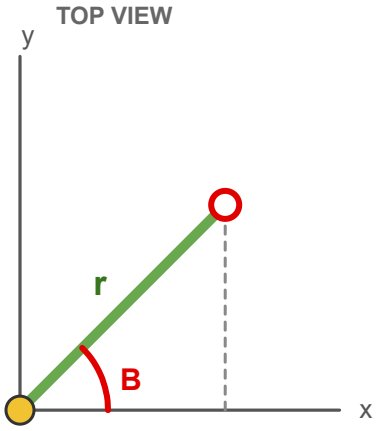
Find B, S and E to put the tip at the point from Lesson 4.

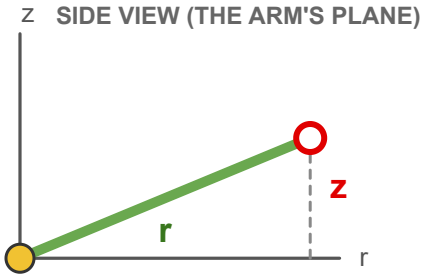
GIVEN

(x, y, z) = (301.78, 301.78, 176.78)

$L_1 = L_2 = 250$ mm

We expect **B = S = E = 45°**, because the Lesson 4 example used those angles. Let's see if the triangles agree.





DAY 2 · WORKED EXAMPLE · STEPS 1 AND 2

Worked Example: Aim and Flatten

STEP 1 · THE BASE

$B = \text{atan2}(301.78, 301.78)$

$B = 45°$

STEP 2 · FLATTEN

$r = \sqrt{(301.78^2 + 301.78^2)} = \mathbf{426.78}$

$h = z = \mathbf{176.78}$

Now the problem is the flat arm with target **(X, Y) = (426.78, 176.78)**.

That is exactly the Day 1 worked example, so the rest is already done: E = 45°, S = 45°.

DAY 2 · WORKED EXAMPLE · STEPS 3 AND 4

Worked Example: Elbow and Shoulder

STEP 3 · ELBOW

$$D^2 = 426.78^2 + 176.78^2 = 213{,}389$$

$$\cos E = \frac{213{,}389 - 125{,}000}{125{,}000} = 0.7071 \rightarrow \mathbf{E = 45°}$$

STEP 4 · SHOULDER

$$\varphi = \tan^{-1}(176.78 \div 426.78) = 22.5°, \quad \cos \delta = \frac{213{,}389}{230{,}970} = 0.9239 \rightarrow \delta = 22.5°$$

$$\mathbf{S = 90° - (22.5° + 22.5°) = 45°}$$

ANSWER **B = 45°, S = 45°, E = 45°**

DAY 2 · CHECK

Check It with Lesson 4

TARGET
(301.78, 301.78, 176.78)

→

IK (LESSON 6)
B, S, E = 45°, 45°, 45°

→

FK (LESSON 4)
(301.78, 301.78, 176.78)

A good habit: **always check an IK answer with FK.** If FK lands back on the target, the angles are right.

JOINT	θ	d	a	α
1 · Base	B = 45°	0	0	90°
2 · Shoulder	90° − 45° = 45°	0	250	0°
3 · Elbow	−45°	0	250	0°

DAY 2 · TRY IT YOURSELF

Try It Yourself #4

Find B, S and E for the target **(187.50, 324.76, 216.51)**.

$L_1 = L_2 = 250$ mm

SOLUTION

B = atan2(324.76, 187.50) = 60°

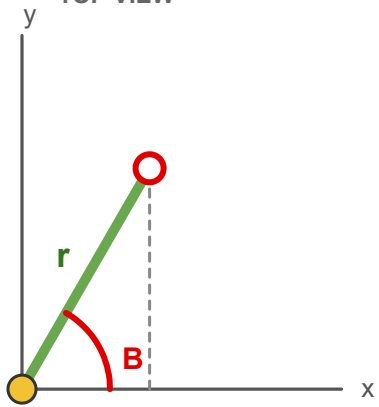
r = 375.00, h = 216.51, D^2 = 187,500

cos E = (187,500 − 125,000) ÷ 125,000 = 0.5, so E = 60°

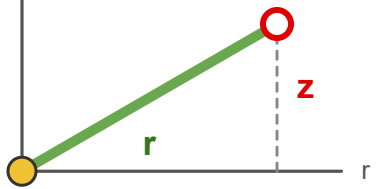
φ = $\tan^{-1}$(216.51 ÷ 375) = 30°, δ = 30°

B = 60°, S = 90° − 60° = 30°, E = 60°

TOP VIEW



z SIDE VIEW (THE ARM'S PLANE)



DAY 2 · TRY IT YOURSELF

Try It Yourself #5

Find B, S and E for the target **(−324.76, 187.50, 216.51)**.

$L_1 = L_2 = 250$ mm

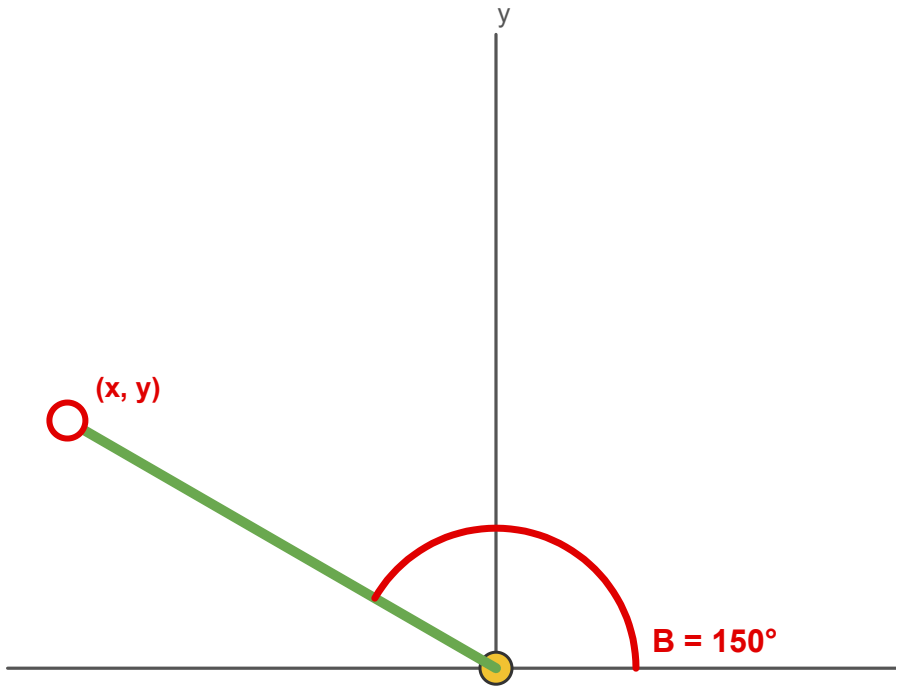
SOLUTION

B = atan2(187.50, −324.76) = 150° (plain $\tan^{-1}$ says −30°)

r = 375.00, h = 216.51, the same as #4

B = 150°, S = 30°, E = 60°

Same arm shape as #4, with the base turned 90° further.



DAY 2 · TRY IT YOURSELF

Try It Yourself #6

Can the arm reach each target? Check D against 500 mm.

(a) (200, 200, 200) **(b) (400, 300, 200)**

SOLUTION

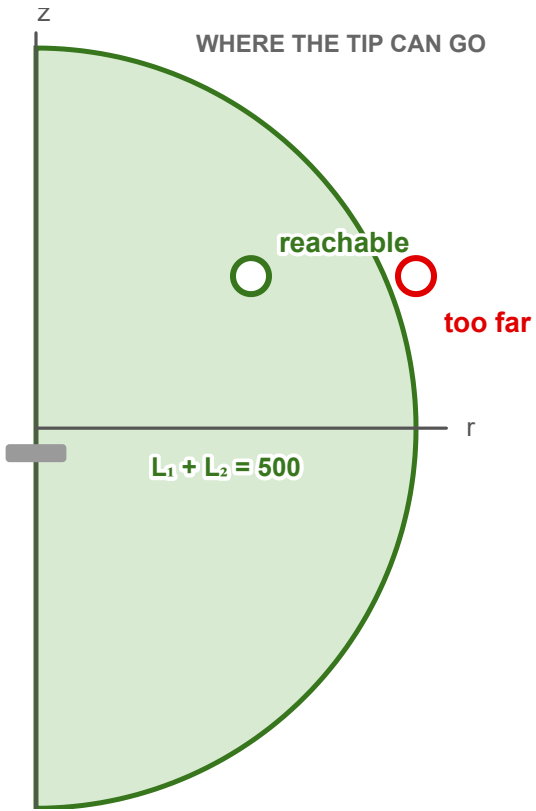
(a) r = 282.84, h = 200, D = 346.41 ≤ 500

Reachable

(b) r = 500, h = 200, D = 538.52 > 500

Out of reach: cos E = 1.32, impossible

For (b) the robot should report an error instead of moving.



DAY 2 · CHECK WITH CODE

Check It with Code

```
from ik_helper import ik_trig
from dh_helper import dh_matrix, position

B, S, E = ik_trig(301.78, 301.78, 176.78)
print(round(B, 2), round(S, 2), round(E, 2))

# round trip through Lesson 4's FK
A1 = dh_matrix(B, 0, 0, 90)
A2 = dh_matrix(90 - S, 0, 250, 0)
A3 = dh_matrix(-E, 0, 250, 0)
print(position(A1 @ A2 @ A3))
```

OUTPUT

```
45.0 45.0 45.0
[301.78, 301.78, 176.78]
```

The first line is the IK answer. The second is the FK check. It lands back on the target.

CHECK YOUR ANSWERS

Change the target on line 4 to the values from Try It Yourself #4 and #5. Then try #6(b) and read the error.

WHAT EACH PART DOES

ik_trig follows the four-step recipe. arctan2 and arccos do the inverse trig. Out-of-reach targets raise an error.

DAY 2 · EDGE CASES

When IK Gets Tricky

FK always gives one answer. IK can give a different number of answers:

CASE	ANSWERS	WHY
D > 500 mm	**none**	too far, the triangle cannot close
D = 500 mm	**one**	arm fully straight, E = 0°
Inside the circle	**two**	elbow up or elbow down
x = y = 0 (on the base axis)	**infinite**	any B works, atan2(0, 0) is undefined

Near full reach, a tiny change in the target changes E a lot, so rounding matters. The code allows a tiny tolerance.

DAY 2 · COMPARISON

Trigonometry for IK: Pros and Cons

PROS

+ An exact answer with a few formulas, no repeating
+ Fast, so a tiny computer can solve it
+ You can see the triangles
+ Easy to check with Lesson 4's FK

CONS

− New triangles for every new joint, like Lesson 3
− Many special cases: quadrants, reach, two solutions
− Only works for this arm's shape

If the arm had six joints, which method would you use for IK? Why?

Trigonometry gives a clean answer for a simple arm like ours. Bigger arms need a method that does not depend on drawing new triangles.

DAY 2 · WRAP-UP

What We Learned

1. IK goes from **target** to **angles**. It may have 0, 1 or 2 answers.
2. The arm makes a **triangle**, so the **law of cosines** gives E and δ.
3. B = **atan2(y, x)**, and r = $\sqrt{(x^2 + y^2)}$ flattens 3D into 2D.
4. S = 90° − (φ + δ). Always **check the result with FK**.

NEXT TIME · LESSON 7

The Gradient Descent Approach to IK

Trigonometry works for our arm. Next we guess, measure the error and improve, a method that works for any arm.